

CIE-1 Question Paper1

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1	Consider the following code of Singly Linked List: <pre> void prints(struct Node* head) { struct Node* current = head; int index = 1; while (current != NULL) { if (index % 2 == 1) { printf("%d ", current->data); } current = current->next; index++; } printf("\n"); } </pre> Input : 11->22->33->44->55->66 A. 11->22->55->66 B. 11->22->33->55 C. 11->33->44->55 D. 11->33->55->66 Answer: D
2	If a node 'X' comprises of three address as 2070, 4250, 7300. Relate the mentioned address as Node Address, Next Node address and Previous Node Address. Find the next pointer address of 'X's previous node. A. 2070 B. 4250 C. 7300 D. 8400 Answer : A
3	The following C function takes a single-linked list of integers as a parameter and rearranges the elements of the list. The function is called with the list containing the integers 1,2,3,4,5,6,7 in the given order. What will be the contents of the list after the function completes execution? struct node {

	<pre> int value; struct node *next; }; void rearrange (struct node *list){ struct node *p, * q; int temp; if(!list !list-> next) return; p = list; q = list->next; while (q) { temp = p->value; p-> value = q ->value; q-> value = temp; p = q-> next; q = p ? p->next : 0; } } </pre> A. 1,2,3,4,5,6,7 B. 2,1,4,3,6,5,7 C. 1,3,2,5,4,7,6 D. 2,3,4,5,6,7,1 Answer :
4	Assume a Doubly Linked List whose data elements are 59,68,78,32,45. Every node address is incremented by 2000 by fixing 700 as its starting address. Delete '32' from the list. Find the previous pointer of '78'. (a) 4700 (b) 8700 (c) 700 (d) 2700 Answer: (d) 2700
5	Consider a two dimensional array A[20][10]. Assume 4 words per memory cell, the base address of array A is 100, elements are stored in row-major order and first element is A[0][0]. What is the address of A[11][5]? (a) 560 (b) 460 (c) 570 (d) 575 Answer: (a) 560
6	Consider the following concept for Linked List procedure: <pre> P=create(node); Info(p) = 10; </pre>

	Next(P) = NULL; Next(List)=P; List=P; A. POP operation B. Removal of a node C. Appending a node D. Modifying an existing node Answer : C. Appending a node
7	What is the output printed by the following program? <pre>#include <stdio.h> int f(int n, int k) { if(n==0) return 0; else if(n%2) return f(n/2, 2*k)+k; else return f(n/2, 2*k)-k; } int main() { printf("%d", f(20,1)); return 0; }</pre> (a) 5 (b) 8 (c) 9 (e) 20 Answer: C. 9
8	Find the Output of the given 'C' code: <pre>void fun(struct node* start) { if(start == NULL) return; printf("%d ", start->data); if(start->next != NULL) fun(start->next->next); printf("%d ", start->data); }</pre> (a) 1 4 6 6 4 1 (b) 1 3 5 1 3 5 (c) 1 2 3 5 (d) 1 3 5 5 3 1 Answer: D. 1 3 5 5 3 1

9	Consider the following C function: <pre> int f(int n) { static int r=0; if(n<=0) return 1; if(n>3) { r=n; return f(n-2)+2; } return f(n-1)+r; } </pre> What is the value of f(5)? (a) 5 (b) 7 (c) 9 (d) 18 Answer : (d) 18
10	Consider the code fragment written in C below: <pre> void f(int n) { if(n<=1) { printf("%d",n); } else { f(n/2); printf("%d", n%2); } } </pre> What does f(173) print? (a) 010110101 (b) 010101101 (c) 10110101 (d) 10101101 Answer : D.10101101

11	For the given 'C' Code <pre>#include <stdio.h> int main() { int arr[3] = {1, 2, 3, 4}; printf("%d", arr[0]); return 0; }</pre> Which of the following describes the error? a) The array initialization has too many elements for the declared size. b) The printf statement incorrectly accesses the array. c) The array should be declared as <code>int arr[]</code> for proper initialization. d) There is no error; the code is valid. Answer: a
12	Which of the following will occur if we call the free() function on a NULL pointer? a) Compilation Error. b) Runtime Error. c) Undefined Behaviour. d) The program will execute normally. Answer: (d) The program will execute normally.
13	<pre>#include <stdio.h> #include<stdlib.h> void set(int *to) { to = (int*)malloc(5 * sizeof(int)); } void solve() { int *ptr; set(ptr); *ptr = 10; printf("%d", *ptr); } int main() { solve(); return 0; }</pre> (a) 10 (b) Garbage value (c) Not determined (d) The program may crash Answer: The program may crash
14	Consider the following program, where are i, j and k are stored in memory? <pre>#include <stdio.h> #include <stdlib.h></pre>

	<pre>int i; int main() { int j; int *k = (int *) malloc (sizeof(int)); }</pre> A. i, j and *k are stored in stack segment B. i and j are stored in stack segment. *k is stored on heap. C. i is stored in BSS part of data segment, j is stored in stack segment. *k is stored on heap. D. j is stored in BSS part of data segment, i is stored in stack segment. *k is stored on heap. Ans : C
15	Local variables are stored in an area called _____ A. Heap B. Permanent storage area C. Free memory D. Stack Ans : D
16	What is the return type of malloc() or calloc()? A. int * B. int ** C. void * D. void ** Ans : C void *
17	Consider the following C program. <pre>#include < stdio.h > int main () { int a [4] [5] = {{1, 2, 3, 4, 5}, {6, 7, 8, 9, 10}, {11, 12, 13, 14, 15}, {16, 17, 18, 19, 20}}; printf ("%d\n", *(a+**a+2) +3)); return (0); }</pre> The Output is (a) 9 (b) 19 (b) -9 (d) 20 Answer : (b) 19
18	Which of the following statements about circular linked lists is FALSE?

	A. Last node points to the first node B. There is no NULL pointer in the list C. Traversal must stop when we reach NULL D. The list can be traversed from any node Answer: C. Traversal must stop when we reach NULL
19	The following C function takes a singly-linked list as input argument. It modifies the list by moving the last element to the front of the list and returns the modified list. A part of the code is left blank. <pre> struct node { int value; struct node *next; }; struct node *move_to_front(struct node *head) { struct node *p, *q; if ((head == NULL) (head->next == NULL)) return head; q = NULL; p = head; while (p->next != NULL) { q = p; p = p->next; } _____ return head; } </pre> Choose the correct alternative to replace the blank line. A) q = NULL; p->next = head; head = p; B) q->next = NULL; head = p; p->next = head; C) head = p; p->next = q; q->next = NULL; D) q->next = NULL; p->next = head; head = p; Answer : D
20	In the following code <pre> struct Node { </pre>

	<pre>int quiz; struct Node* test; }; int main() { struct Node certification, assignment, exam; ----- }</pre> Memory for the struct objects is created _____ - a)on stack - b)on heap - c)both on stack and heap - d)neither stack nor heap Answer : a) on stack
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